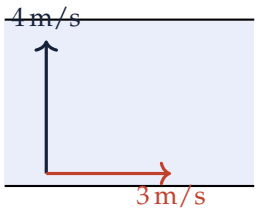
AP PHYSICS 1

Unit 1 • Lesson 1.4 • Guided notes

Topic 1.4 • Reference Frames and Relative Motion

Guided notes • Fill in the blanks and work the examples in the boxes

Objective: describe motion from different reference frames and add relative velocities.

Reference frames	Position and velocity depend on the _____: the reference frame. A passenger sitting in a moving bus is _____ relative to the bus and _____ relative to the road. Two frames moving at constant velocity relative to each other measure the _____.
Adding relative velocities	$\vec{v}_{A \text{ rel } C} =$ _____ A train moves at 20 m/s east. A passenger walks at 1.5 m/s relative to the train. Walking east: _____ east relative to the ground. Walking west: _____ east relative to the ground.
Example: crossing a river	A boat points straight across a river 80 m wide at 4 m/s relative to the water. The water flows at 3 m/s. Speed relative to the bank: _____ Time to cross: _____ (only the _____ velocity matters) Drift downstream: _____ 
Check	Car A drives at 25 m/s east, car B at 15 m/s east. Velocity of A relative to B: _____. If B drives west at 15 m/s: _____.
Summary	